

1) b) Data and a pointer to the next node

2) c) NULL

3) b) Self referential structure

4) b) head == NULL

5) b) newnode $\rightarrow$ next = head ;
head = new node

6) b) temp = head; head = head $\rightarrow$
next; free (temp);

7) c) Traversing from tail to head

8) b) n

9) c) Copying the data of p $\rightarrow$
next into p, then deleting p $\rightarrow$
next

10) b) while (p != NULL)

11) b) p $\rightarrow$ next == NULL

12) c) two pointers advancing at
different rates

13) c) Three

14) b) Exactly one node.

15) b) p = (struct node *) malloc
(size of (struct node));

16) b) Avoid special-case handling
for insertion and deletion of
the head.

17) c) the first node

18) b) It does not support random
(indexed) access

19) b) Loss of access to all nodes,
causing a memory leak.

20) b) A slow pointer and a fast
pointer

21) c) The last node

22) b) Nodes may be scattered
anywhere in the memory.

23) struct node {
 int data;
 struct node * next;
};

24) 305070

28) i) 1 $\rightarrow$ 5 $\rightarrow$ 8 $\rightarrow$ 12 $\rightarrow$ 3 $\rightarrow$ NULL

ii) 7 $\rightarrow$ 5 $\rightarrow$ 8 $\rightarrow$ 8 $\rightarrow$ NULL

iii) 7 $\rightarrow$ 5 $\rightarrow$ 8 $\rightarrow$ 3 $\rightarrow$ 9 $\rightarrow$ NULL